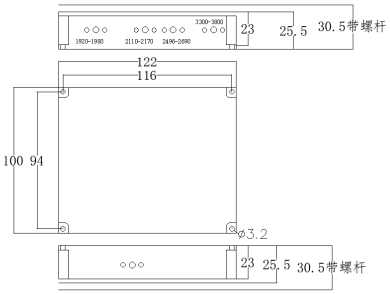
四端口腔体合路器

KXT-CM4-B1N78 Band 1 + Band 41/n41 + n78

项目	指标
CH1 工作频率	1920–1980 MHz
CH1 带宽	60 MHz
CH1 插入损耗	≤ 1.8 dB
CH1 带内波动	≤ 1.5 dB
CH1 端口驻波	≤ 1.3
CH1 回波损耗	≥ 17.7 dB（由驻波换算）
CH2 工作频率	2110–2170 MHz
CH2 带宽	60 MHz
CH2 插入损耗	≤ 1.8 dB
CH2 带内波动	≤ 1.5 dB
CH2 端口驻波	≤ 1.3
CH2 回波损耗	≥ 17.7 dB（由驻波换算）
CH3 工作频率	2496–2690 MHz
CH3 带宽	194 MHz
CH3 插入损耗	≤ 1.8 dB
CH3 带内波动	≤ 1.5 dB
CH3 端口驻波	≤ 1.3
CH3 回波损耗	≥ 17.7 dB（由驻波换算）
CH4 工作频率	3300–3800 MHz
CH4 带宽	500 MHz
CH4 插入损耗	≤ 1.8 dB
CH4 带内波动	≤ 1.5 dB
CH4 端口驻波	≤ 1.3
CH4 回波损耗	≥ 17.7 dB（由驻波换算）
功率容量	20 W 平均
接头	SMA，公母下单确认
工作温度	−20 °C ~ +60 °C
外形尺寸	122 × 100 × 30.5 mm
外观	铝合金腔体。图纸标注高度 30.5 mm，含调谐螺杆。



带外抑制 / 端口隔离

端口	频段或频点	指标
CH1	2110–2170 MHz	≥ 80 dB
CH1	2496–2690 MHz	≥ 80 dB
CH1	3300–3800 MHz	≥ 80 dB
CH2	1920–1980 MHz	≥ 80 dB
CH2	2496–2690 MHz	≥ 80 dB
CH2	3300–3800 MHz	≥ 80 dB
CH3	1920–1980 MHz	≥ 80 dB
CH3	2110–2170 MHz	⩾ 80 dB

CH3	3300–3800 MHz	≥ 80 dB
CH4	1920–1980 MHz	≥ 80 dB
CH4	2110–2170 MHz	⩾ 80 dB
CH4	2496–2690 MHz	≥ 80 dB

本系列统一规范（订货与出厂）

特性阻抗	50 Ω
使用场合	室内型
腔体	铝合金，导电氧化
测试条件	25 °C，50 Ω，矢量网络分析仪
低互调订货档	PIM3 ≤ −150 dBc @ 2×43 dBm
湿度	0 ~ 95% RH
储存温度	−20 °C ~ +60 °C
机械	安装孔距 116 × 94 mm，孔径 Ø3.2 mm。

说明

- 回波损耗由端口驻波比 ⩽ 1.3 换算：RL = −20 log10((VSWR−1)/(VSWR+1)) ≈ 17.7 dB。
- 低互调档是订货选项。原稿没有给出实测互调，未选该档时不作为本页保证值。需要时按 2×43 dBm 测试并附报告。
- CH3、CH4 原稿把 1920–1980 MHz 写了两遍，漏了 2110–2170 MHz；隔离按同表其余端口补为 ⩾80 dB。
- 规格来源：四端口合路器规格7.15.docx

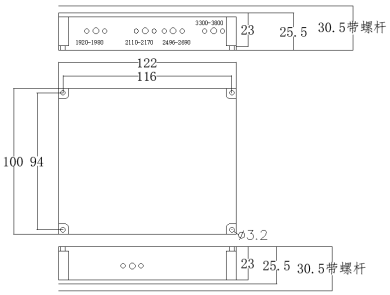
订货写法：KXT-CM4-B1N78 + 接头公母 + 颜色（本色/黑色）+ 是否低互调档 + 数量。

English

4-port Cavity Combiner

KXT-CM4-B1N78 Band 1 + Band 41/n41 + n78

Item	Specification
CH1 Passband	1920–1980 MHz
CH1 Bandwidth	60 MHz
CH1 Insertion loss	≤ 1.8 dB
CH1 Passband ripple	≤ 1.5 dB
CH1 VSWR	≤ 1.3
CH1 Return loss	≥ 17.7 dB (from VSWR)
CH2 Passband	2110–2170 MHz
CH2 Bandwidth	60 MHz
CH2 Insertion loss	≤ 1.8 dB
CH2 Passband ripple	≤ 1.5 dB
CH2 VSWR	≤ 1.3
CH2 Return loss	≥ 17.7 dB (from VSWR)
CH3 Passband	2496–2690 MHz
CH3 Bandwidth	194 MHz
CH3 Insertion loss	≤ 1.8 dB
CH3 Passband ripple	≤ 1.5 dB
CH3 VSWR	≤ 1.3
CH3 Return loss	≥ 17.7 dB (from VSWR)
CH4 Passband	3300–3800 MHz
CH4 Bandwidth	500 MHz
CH4 Insertion loss	≤ 1.8 dB
CH4 Passband ripple	≤ 1.5 dB
CH4 VSWR	≤ 1.3
CH4 Return loss	≥ 17.7 dB (from VSWR)
Power handling	20 W average
Connector	SMA, confirm gender on the order
Operating temperature	−20 °C ~ +60 °C
Outline	122 × 100 × 30.5 mm
Finish	Aluminum cavity. Drawing height is 30.5 mm including tuning screws.



Rejection / port isolation

Port	Band or point	Limit
CH1	2110–2170 MHz	≥ 80 dB
CH1	2496–2690 MHz	≥ 80 dB
CH1	3300–3800 MHz	≥ 80 dB
CH2	1920–1980 MHz	≥ 80 dB
CH2	2496–2690 MHz	≥ 80 dB
CH2	3300–3800 MHz	≥ 80 dB
CH3	1920–1980 MHz	≥ 80 dB

CH3	2110–2170 MHz	⩾ 80 dB
CH3	3300–3800 MHz	≥ 80 dB
CH4	1920–1980 MHz	≥ 80 dB
CH4	2110–2170 MHz	⩾ 80 dB
CH4	2496–2690 MHz	≥ 80 dB

Series standard (order and shipment)

Impedance	50 Ω
Use	Indoor
Construction	Aluminum alloy, conductive oxidation
Test condition	25 °C, 50 Ω, vector network analyzer
Low-PIM order option	PIM3 ≤ −150 dBc @ 2×43 dBm
Humidity	0 ~ 95% RH
Storage	−20 °C ~ +60 °C
Mechanical	Mounting-hole pitch 116 × 94 mm, hole Ø3.2 mm.

Notes

- Return loss is calculated from VSWR ⩽ 1.3: RL = −20 log10((VSWR−1)/(VSWR+1)) ≈ 17.7 dB.
- Low PIM is an order option. The source sheet has no measured PIM, so it is not a guaranteed value unless that option is ordered and reported.
- On CH3 and CH4 the source repeated 1920–1980 MHz and omitted 2110–2170 MHz; isolation is filled as ⩾80 dB to match the rest of the sheet.
- Source specification: 四端口合路器规格7.15.docx

Order as: KXT-CM4-B1N78 + connector gender + finish (natural/black) + low-PIM option + quantity.